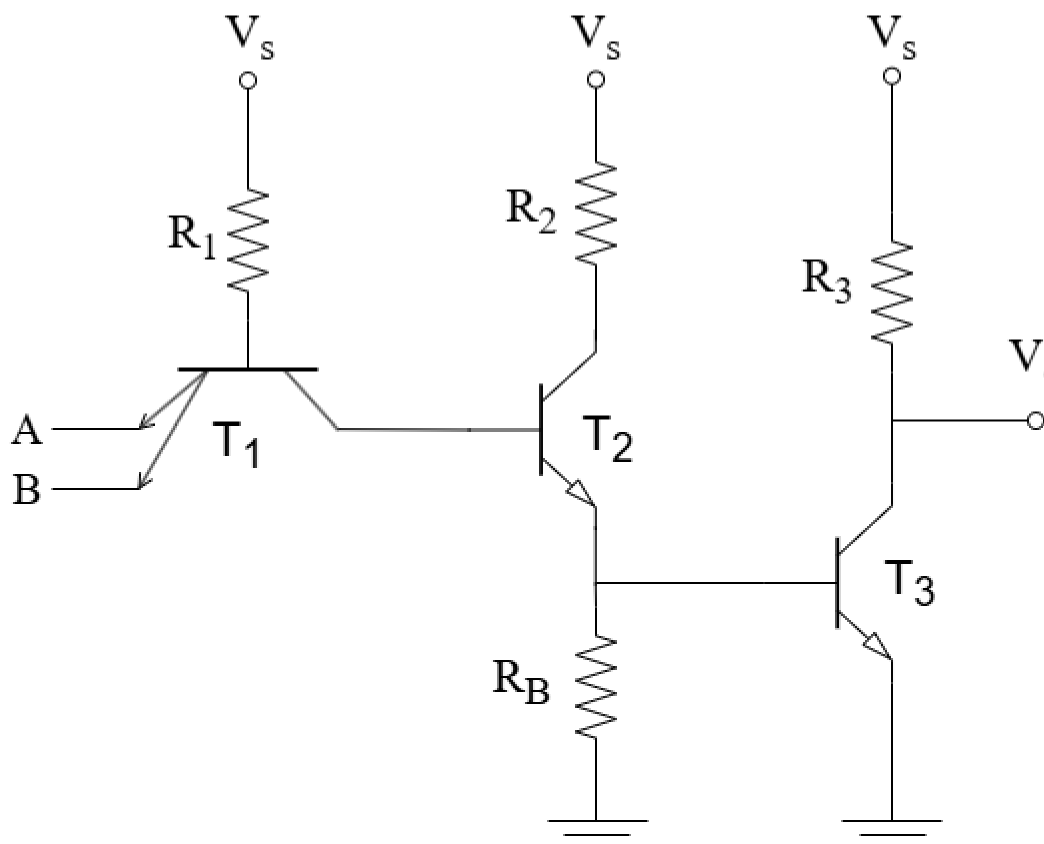
Assignment 2:

Use $V\gamma$(diode) = 0.6 V, VD(conducting voltage of diode) = 0.7 V, $V\gamma$(transistor) = 0.5 V, V_{BE}(forward active) = 0.7 V, V_{BE} (saturation) = 0.8 V, V_{CE} (saturation) = 0.2 V , β_F =30, β_R =0.1, V_{BC} (reverse active) = 0.7 V for all the questions unless otherwise stated.



Given, $V_S = 9$ V , $R_2 = 2$k , $R_3 = 4.4$k, $R_B = 4.4$k

Question 1: Determine the power dissipation of this TTL circuit when both inputs are HIGH. You must show your calculation. Assume $R_1 = 6.7$k for this question only

Question 2: Suppose you are selling this TTL logic gate to a customer and he states a strict criteria of keeping the power dissipation below 1.23 mW when both inputs are LOW (0 V). Determine the maximum value of R_1 you can use to satisfy this criteria.